

Daniel Oppenheimer

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Education

- **University of Virginia** **Aug 2023 – May 2027**
- BS in Computer Science, Data Science Minor, Engineering Business Minor
- Cumulative GPA: 3.78
- Relevant Coursework: Cloud Computing, Parallel Computing, Machine Learning, Artificial Intelligence, Computer Networks, Natural Language Processing (NLP), Software Development Essentials, Software Engineering, Data Structures and Algorithms I & II, Computer Systems and Organization I & II, Discrete Mathematics I & II, New Product Development, Data Science Systems
- Activities and Societies: Trigon Engineering Society @ UVA, Sigma Alpha Mu – Special Events Chair, 2025 Deloitte National Undergraduate Case Competition, Intramural Sports Supervisor

Experience

Comcast, *Software Engineer Intern* **June 2026 – August 2026**

- Migrating Xfinity's customer-facing website repository's CI/CD pipeline from Concourse to GitHub Actions, eliminating hour-long delays per pull request & streamlining releases for a service for 30 million customers.

Booz Allen Hamilton, *AI Software Engineer Intern* **June 2025 – August 2025**

- Engineered and benchmarked three distinct Retrieval-Augmented Generation (RAG) frameworks (Keyword, Hybrid, and Graph RAG) for an AI search tool, directly collaborating with NOAA developers to choose the optimal framework to streamline information retrieval from a complex data storage system.
- Implemented a novel hybrid search system integrating BM25 keyword ranking with semantic search, leveraging a Reciprocal Rank Fusion pipeline and a genetic algorithm to optimize search result relevancy.
- Developed a full-stack solution with a Flask frontend, connecting to a backend on AWS EC2. Used REST API strategies to connect to NOAA data, a Redis cache, and the Gemma3 GenAI model.
- Developed CoT system to auto-upload new data formats in a metadata-rich format.

Zbooni, *Software Development Consultant* **Dec 2025 – Jan 2026**

- Engineered a centralized fraud detection dashboard to automate risk assessment for the compliance team, replacing manual oversight with robust security rules, saving KYC team members 3 hours per workday.
- Architected an automated email verification workflow using Trengo for suspicious transactions, streamlining customer confirmation responses and significantly reducing the platform's exposure to fraudulent activity.
- Prototyped Random Forest and XGBoost models for risk data collected through platform in 6 months.

Glass House Recycling LLC, *Co-founder* **Feb 2020 – August 2025**

- Founded curbside glass-pickup business at age 14 to meet demand after Falls Church City stopped providing municipal glass recycling service.
- Developed in-house invoicing system with a Node.js express web application.
- Leveraged digital advertising and print/radio media (NBC, WAMU, NowThis) to grow customer base to over 160 residential households, online stories about business reached 550k+ views, with 5k+ reactions.
- \$10k+ exit, selling company and converting over 80% of customers to the new platform.

Projects

Drawtex, *Developer* **Dec 2025 – Present**

- Designed AI driven hand-drawn notes app that classifies pen strokes between text, math, and diagrams using a Bi-Directional LSTM model along with advanced feature engineering to achieve an F1 score of 99%.
- Developed advanced converter that formats classified text into Latex, enhancing students' learning workflow.

Bracket Hub, *Developer* **Apr 2024 – Nov 2025**

- Built social/sporting web application to bring a March Madness point-scoring system to user-made tournament brackets.
- React.js that interfaced with Firebase authentication and Firestore database for backend and data storage.

Spyfall, *Developer* **Jan 2020 – Apr 2020**

- Designed minimalist reimagination of the "Spyfall" game front-end webpage using Node.js/HTML/CSS.
- Implemented WebSocket communication using the Socket.io API for live multiplayer capability.

Skills

- Python, JavaScript (React, Node.js, Typescript), Java, HTML/CSS, , C, MySQL, SQLite
- Git, AWS, NPM, PyTorch, RAG, Linux, GitHub Actions, Concourse, Docker, Ollama, Firebase, Flask, Django, REST API, MATLAB, Agile Method, Jupyter Notebook, Blender

Awards: Eagle Scout